IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

RADIO FREQUENCY SUBSYSTEMS

Wednesday 10 May 10:00 am
Time allowed: 3:00 hours

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : S. Lucyszyn
Second Marker(s) : W.T. Pike

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Special instructions for invigilators

This is an OPEN BOOK examination.

Special instructions for students

There are 4 (four) questions in total.

Question 1 is COMPULSORY and carries a weighting of 50%.

Answer 2 (two) out of the remaining 3(three) questions, each carrying a weighting of 25%.

All variables and acronyms have their usual meaning.

1. This question relates to radiometry for passive millimetre-wave imaging.

a)

i) Briefly describe the basic concept behind a passive radiometric imager and state how this differs from a traditional optical camera.

[3]

ii) State a suitable millimetre-wave frequency for operating a passive radiometer for outdoor security applications and briefly explain your choice of frequency.

[3]

iii) Briefly explain the relationship between millimetre-wave frequency and cost for both passive components and active devices.

[2]

iv) With a body scanning radiometer, state examples of the types of materials that can be detected beneath a person's clothing?

[2]

v) Give two examples of satellite-based applications for passive radiometry.

[2]

b) A badly designed radiometric receiver could be just a passive band-pass filter. At its input is an ideal antenna and at its output is a power meter. The relevant specifications for this system are:

Frequency Range: 140 to 220 GHz; Radiometric Receiver Gain: -3 dB;
Radiometric Receiver Ambient Temperature of Operation: 23°C.

Sketch a basic illustration of the complete radiometric system and show all relevant parameters (specifications and variables used in subsequent calculations). Make sure to clearly define all parameters and their units.

[10]

c) The Boltzmann constant $k = 1.380649 \times 10^{-23}$ [J/K] and temperature conversion from Celsius to Absolute T [K] = 273.15 + T [°C]. The radiometric system in 1(b) is used to observe the Sun, having a surface temperature of 5,778 K. With information in 1(b), calculate:

i) Radiometer system bandwidth.

[2]

ii) Radiometric receiver noise figure.

[2]

iii) Input noise power from the antenna, in dBm.

[3]

iv) Radiometric receiver equivalent noise temperature.

[4]

v) Radiometric receiver internal noise power, in dBm.

[2]

vi) Radiometer system equivalent noise temperature.

[2]

vii) Radiometer system noise power, in dBm.

[2]

viii) Output noise power into the power meter, in dBm.

[4]

ix) Required sensitivity for the power meter with 50 Ω input impedance.

[4]

d) Assuming that the noise figure in 1(c) does not change, how can the radiometric receiver be simply redesigned so that it can be used with a power meter having a minimum detectable power level of -25 dBm?

[3]

2.

a)

(i) For a lossless linear two-port network, state the simple equation that shows the relationship between insertion loss and return loss.

[3]

(ii) Using the equation stated in 2(a)(i), calculate the return loss when insertion loss is 1 dB.

[3]

b) Draw a 3-channel frequency de-multiplexer and a 3-channel frequency multiplexer, employing circulators and band-pass filters.

[7]

c) Employing the topologies given in 2(b), draw a unilateral 3-channel amplifier. Assume that for each frequency band the associated filters and amplifiers have the same -1 dB band edge frequencies.

[4]

d) Employing the topology from in 2(c), simple power loss budget analysis can be performed to calculate the overall gain, given the insertion loss of each circulator is 0.1 dB, the mid-band insertion loss of the filters is 0 dB and the mid-band gain of the amplifiers is 10 dB. State any simplifying assumptions.

(i) Calculate the gain at mid-band, assuming that the transition frequency bandwidth is insignificant.

[3]

(ii) Calculate the gain at the -1 dB frequency band edge, assuming that the ideal upper -1 dB band edge frequency for one channel is the same as the ideal lower -1 dB band edge frequency for the adjacent channel.

[3]

(iii) Using the results from 2(d)(i) and (ii), comment on the resulting frequency response over the entire bandwidth of the 3 channels.

[2]

3. A MESFET has input and output impedances of $2 - j\,5\ \Omega$ and $40 - j\,10\ \Omega$, respectively. You are required to design the inter-stage impedance matching network for a two-stage 1 GHz amplifier. Assume that the MESFET input and output impedances are fixed and ignore DC biasing networks.

a) Draw the *RC* lumped-element models for the output of the first-stage transistor and the input for the second-stage transistor. Identify the dominant intrinsic parasitic resistive and capacitive elements of the MESFET.

[4]

b) From the *RC* lumped-element models drawn in 3(a), calculate:

i) Parasitic values for both resistances and both capacitances.

[5]

ii) Using the values in 3(b)(i), calculate the loaded quality factors for the input and output of the MESFET. State if it is the input or output that limits the bandwidth of operation.

[4]

c) Using discrete lumped-element matching, calculate the:

i) lumped-element component values and state their orientation needed to resonate out the intrinsic parasitic capacitances of the MESFET at the output of the first MESFET and input of the second MESFET.

[2]

ii) loaded quality factor of the inter-stage impedance matching network and its bandwidth. State if it is the input of the first MESFET, inter-stage impedance matching network or output of the second MESFET that limits the bandwidth of operation.

[2]

iii) lumped-element component values needed to implement a one-stage resistive L-matching network. Draw the complete impedance matching network having the minimal number of lumped-element components.

[4]

d) With a two-stage resistive L-matching network calculate the new loaded quality factor. State if it is the input of the first MESFET, new inter-stage impedance matching network or output of the second MESFET that limits the bandwidth of operation.

[4]

4.

a) Explain the difference between a duplexer and a diplexer.

[2]

b) What is the most common way of implementing a duplexer in half-duplex and full-duplex mobile phone handsets?

[4]

c) Using the ideal lumped-element *LC* filter synthesis technique to implement a high pass filter and its associated low-pass prototype, design a simple 50 Ω GSM duplexer that can allow the following bands to pass through the duplexer with a 1 dB worst-case pass band attenuation ripple:

TX band: 930 ~ 943 MHz

RX band: 885 ~ 889 MHz

Draw the block diagram for this basic duplexer, indicating the directions of signal flow and any relevant impedances.

Hint: Using the attached Standard Filter Curves and Tables, choose 7th-order filters with a common cut-off frequency and common frequency ratio, giving approximately 10 dB rejection at the nearest band edge of the other band, and ensure that the one port impedance is double that of the other (e.g. the source termination impedance is Zs = 2Zo and the load termination impedance is Z_L = Zo = 50 Ω).

[12]

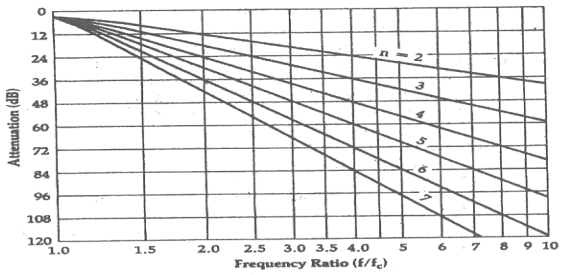
d) Give four reasons why the design in 4(c) is not suitable for a real GSM mobile phone handset duplexer.

[4]

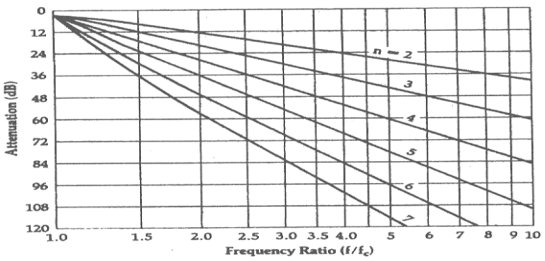
e) Calculate ideal worst-case levels of return loss ripple in the pass bands for the ideal filters used in 4(c).

[3]

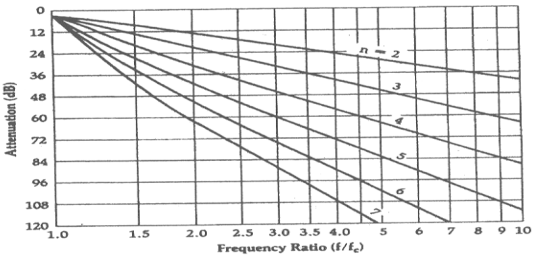
Standard Filter Curves and Tables



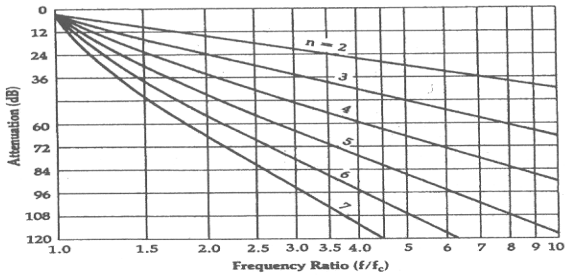
Attenuation characteristics for Butterworth filters.



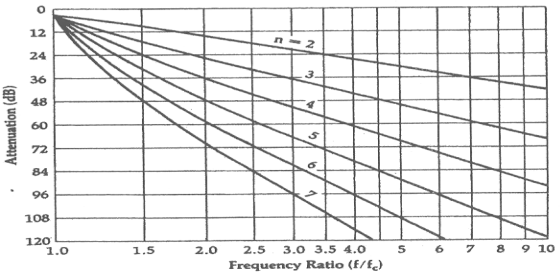
Attenuation characteristics for a Chebyshev filter with 0.01-dB ripple.



Attenuation characteristics for a Chebyshev filter with 0.1-dB ripple.

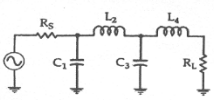


Attenuation characteristics for a Chebyshev filter with 0.5-dB ripple.



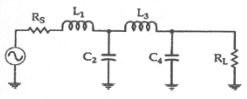
Attenuation characteristics for a Chebyshev filter with 1-dB ripple.

Chebyshev Low-Pass Prototype Element Values for 1.0-dB Ripple

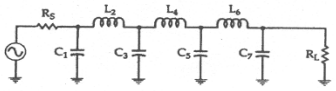


R_S/R_L	C_1	L_2	C_3	L_4
3.000	0.572	3.132		
4.000	0.365	4.600		
8.000	0.157	9.658		
∞	1.213	1.109		
1.000	2.216	1.088	2.216	
0.500	4.431	0.817	2.216	
0.333	6.647	0.726	2.216	
0.250	8.862	0.680	2.216	
0.125	17.725	0.612	2.216	
∞	1.652	1.460	1.108	
3.000	0.653	4.411	0.814	2.535
4.000	0.452	7.083	0.612	2.848
8.000	0.209	17.164	0.428	3.281
∞	1.350	2.010	1.488	1.106

R_L/R_S	L_1	C_2	L_3	C_4

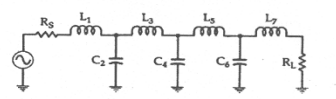


Chebyshev Low-Pass Prototype Element Values for 1.0-dB Ripple

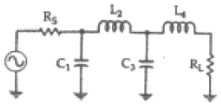


n	R_S/R_L	C_1	L_2	C_3	L_4	C_5	L_6	C_7
5	1.000	2.207	1.128	3.103	1.128	2.207		
	0.500	4.414	0.565	4.653	1.128	2.207		
	0.333	6.622	0.376	6.205	1.128	2.207		
	0.250	8.829	0.282	7.756	1.128	2.207		
	0.125	17.657	0.141	13.961	1.128	2.207		
	∞	1.721	1.645	2.061	1.493	1.103		
6	3.000	0.679	3.873	0.771	4.711	0.969	2.406	
	4.000	0.481	5.644	0.476	7.351	0.849	2.582	
	8.000	0.227	12.310	0.198	16.740	0.726	2.800	
	∞	1.378	2.097	1.690	2.074	1.494	1.102	
7	1.000	2.204	1.131	3.147	1.194	3.147	1.131	2.204
	0.500	4.408	0.566	6.293	0.895	3.147	1.131	2.204
	0.333	6.612	0.377	9.441	0.798	3.147	1.131	2.204
	0.250	8.815	0.283	12.588	0.747	3.147	1.131	2.204
	0.125	17.631	0.141	25.175	0.671	3.147	1.131	2.204
	∞	1.741	1.677	2.155	1.703	2.079	1.494	1.102

n	R_L/R_S	L_1	C_2	L_3	C_4	L_5	C_6	L_7

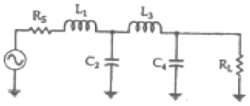


Butterworth Low-Pass Prototype Element Values

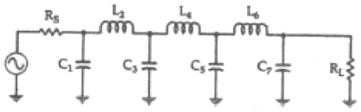


n	R_S/R_L	C_1	L_2	C_3	L_4
2	1.111	1.035	1.835		
	1.250	0.849	2.121		
	1.429	0.697	2.439		
	1.667	0.566	2.828		
	2.000	0.448	3.346		
	2.500	0.342	4.095		
	3.333	0.245	5.313		
	5.000	0.156	7.707		
	10.000	0.074	14.814		
	∞	1.414	0.707		
3	0.900	0.808	1.633	1.599	
	0.800	0.844	1.384	1.926	
	0.700	0.915	1.165	2.277	
	0.600	1.023	0.965	2.702	
	0.500	1.181	0.779	3.261	
	0.400	1.425	0.604	4.064	
	0.300	1.838	0.440	5.383	
	0.200	2.669	0.284	7.910	
	0.100	5.167	0.138	15.455	
	∞	1.500	1.333	0.500	
4	1.111	0.466	1.592	1.744	1.469
	1.250	0.388	1.695	1.511	1.811
	1.429	0.325	1.862	1.291	2.175
	1.667	0.269	2.103	1.062	2.613
	2.000	0.218	2.452	0.883	3.187
	2.500	0.169	2.986	0.691	4.009
	3.333	0.124	3.883	0.507	5.338
	5.000	0.080	5.684	0.331	7.940
	10.000	0.039	11.094	0.162	15.642
	∞	1.531	1.577	1.082	0.383

n	R_L/R_S	L_1	C_2	L_3	C_4

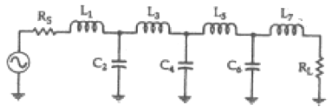


Butterworth Low-Pass Prototype Element Values

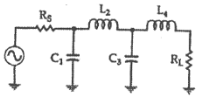


n	R_S/R_L	C_1	L_2	C_3	L_4	C_5	L_6	C_7
5	0.900	0.442	1.027	1.910	1.756	1.389		
	0.800	0.470	0.866	2.061	1.544	1.738		
	0.700	0.517	0.731	2.285	1.333	2.108		
	0.600	0.586	0.609	2.600	1.126	2.552		
	0.500	0.686	0.496	3.051	0.924	3.133		
	0.400	0.838	0.388	3.736	0.727	3.965		
	0.300	1.094	0.285	4.884	0.537	5.307		
	0.200	1.608	0.186	7.185	0.352	7.935		
	0.100	3.512	0.091	14.095	0.173	15.710		
	∞	1.545	1.694	1.382	0.894	0.309		
6	1.111	0.289	1.040	1.322	2.054	1.744	1.335	
	1.250	0.245	1.116	1.126	2.239	1.550	1.688	
	1.429	0.207	1.236	0.957	2.499	1.346	2.062	
	1.667	0.173	1.407	0.801	2.858	1.143	2.509	
	2.000	0.141	1.653	0.654	3.369	0.942	3.094	
	2.500	0.111	2.028	0.514	4.141	0.745	3.931	
	3.333	0.082	2.656	0.379	5.433	0.552	5.280	
	5.000	0.054	3.917	0.248	8.020	0.363	7.922	
	10.000	0.026	7.705	0.122	15.786	0.179	15.738	
	∞	1.553	1.759	1.553	1.202	0.758	0.259	
7	0.900	0.299	0.711	1.404	1.489	2.125	1.727	1.296
	0.800	0.322	0.606	1.517	1.278	2.334	1.546	1.652
	0.700	0.357	0.515	1.688	1.091	2.618	1.350	2.028
	0.600	0.408	0.432	1.928	0.917	3.005	1.150	2.477
	0.500	0.480	0.354	2.273	0.751	3.553	0.951	3.064
	0.400	0.590	0.278	2.795	0.592	4.380	0.754	3.904
	0.300	0.775	0.206	3.671	0.437	5.761	0.560	5.258
	0.200	1.145	0.135	5.427	0.287	8.526	0.369	7.908
	0.100	2.257	0.067	10.700	0.142	16.822	0.182	15.748
	∞	1.558	1.799	1.659	1.397	1.055	0.656	0.223

n	R_L/R_S	L_1	C_2	L_3	C_4	L_5	C_6	L_7

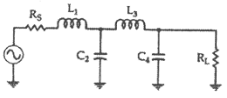


Chebyshev Low-Pass Element Values for 0.01-dB Ripple

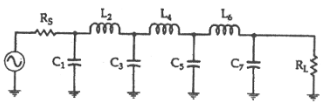


n	R_S/R_L	C_1	L_2	C_3	L_4
2	1.101	1.347	1.483		
	1.111	1.247	1.595		
	1.250	0.943	1.997		
	1.429	0.759	2.344		
	1.667	0.609	2.750		
	2.000	0.479	3.277		
	2.500	0.363	4.033		
	3.333	0.259	5.255		
	5.000	0.164	7.650		
	10.000	0.078	14.749		
	∞	1.412	0.742		
3	1.000	1.181	1.821	1.181	
	0.900	1.092	1.660	1.480	
	0.800	1.097	1.443	1.806	
	0.700	1.160	1.228	2.185	
	0.600	1.274	1.024	2.598	
	0.500	1.452	0.829	3.164	
	0.400	1.734	0.645	3.974	
	0.300	2.216	0.470	5.280	
	0.200	3.193	0.305	7.834	
	0.100	6.141	0.148	15.390	
	∞	1.501	1.433	0.591	
4	1.100	0.950	1.938	1.761	1.046
	1.111	0.854	1.946	1.744	1.165
	1.250	0.618	2.075	1.542	1.617
	1.429	0.495	2.279	1.334	2.008
	1.667	0.398	2.571	1.128	2.461
	2.000	0.316	2.994	0.926	3.045
	2.500	0.242	3.641	0.729	3.875
	3.333	0.174	4.727	0.538	5.209
	5.000	0.112	6.910	0.352	7.813
	10.000	0.054	13.469	0.173	15.510
	∞	1.529	1.694	1.312	0.523

n	R_L/R_S	L_1	C_2	L_3	C_4

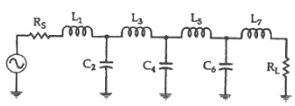


Chebyshev Low-Pass Element Values for 0.01-dB Ripple

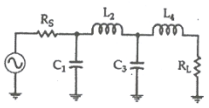


n	R_S/R_L	C_1	L_2	C_3	L_4	C_5	L_6	C_7
5	1.000	0.977	1.685	2.037	1.685	0.977		
	0.900	0.880	1.456	2.174	1.641	1.274		
	0.800	0.877	1.235	2.379	1.499	1.607		
	0.700	0.926	1.040	2.658	1.323	1.977		
	0.600	1.019	0.863	3.041	1.135	2.424		
	0.500	1.166	0.699	3.584	0.942	3.009		
	0.400	1.398	0.544	4.403	0.749	3.845		
	0.300	1.797	0.398	5.772	0.557	5.193		
	0.200	2.604	0.259	8.514	0.368	7.826		
	0.100	5.041	0.127	16.741	0.182	15.613		
	∞	1.547	1.795	1.645	1.237	0.488		
6	1.101	0.851	1.796	1.841	2.027	1.631	0.937	
	1.111	0.760	1.782	1.775	2.094	1.638	1.053	
	1.250	0.545	1.864	1.489	2.403	1.507	1.504	
	1.429	0.436	2.038	1.266	2.735	1.332	1.899	
	1.667	0.351	2.298	1.061	3.167	1.145	2.357	
	2.000	0.279	2.678	0.867	3.768	0.954	2.948	
	2.500	0.214	3.261	0.682	4.667	0.761	3.790	
	3.333	0.155	4.245	0.503	6.163	0.568	5.143	
	5.000	0.100	6.223	0.330	9.151	0.376	7.785	
	10.000	0.048	12.171	0.162	18.105	0.187	15.595	
	∞	1.551	1.847	1.790	1.598	1.190	0.469	
7	1.000	0.913	1.595	2.002	1.870	2.002	1.595	0.913
	0.900	0.816	1.362	2.089	1.722	2.202	1.581	1.206
	0.800	0.811	1.150	2.262	1.525	2.465	1.464	1.538
	0.700	0.857	0.967	2.516	1.323	2.802	1.307	1.910
	0.600	0.943	0.803	2.872	1.124	3.250	1.131	2.359
	0.500	1.080	0.650	3.382	0.928	3.875	0.947	2.948
	0.400	1.297	0.507	4.156	0.735	4.812	0.758	3.790
	0.300	1.669	0.372	5.454	0.546	6.370	0.568	5.148
	0.200	2.242	0.242	8.057	0.360	9.484	0.378	7.802
	0.100	4.701	0.119	15.872	0.178	18.818	0.188	15.652
	∞	1.559	1.867	1.866	1.765	1.563	1.161	0.456

n	R_L/R_S	L_1	C_2	L_3	C_4	L_5	C_6	L_7

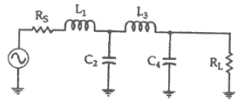


Chebyshev Low-Pass Prototype Element Values for 0.1-dB Ripple

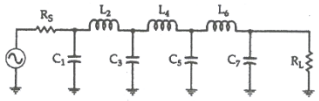


n	R_S/R_L	C_1	L_2	C_3	L_4
2	1.355	1.209	1.638		
	1.429	0.977	1.982		
	1.667	0.733	2.489		
	2.000	0.560	3.054		
	2.500	0.417	3.827		
	3.333	0.293	5.050		
	5.000	0.184	7.426		
	10.000	0.087	14.433		
	∞	1.391	0.819		
3	1.000	1.433	1.594	1.433	
	0.900	1.426	1.494	1.622	
	0.800	1.451	1.356	1.871	
	0.700	1.521	1.193	2.190	
	0.600	1.648	1.017	2.603	
	0.500	1.853	0.838	3.159	
	0.400	2.186	0.660	3.968	
	0.300	2.763	0.486	5.279	
	0.200	3.942	0.317	7.850	
	0.100	7.512	0.155	15.466	
	∞	1.513	1.510	0.716	
4	1.355	0.992	2.148	1.585	1.341
	1.429	0.779	2.348	1.429	1.700
	1.667	0.576	2.730	1.185	2.243
	2.000	0.440	3.227	0.967	2.856
	2.500	0.329	3.961	0.760	3.698
	3.333	0.233	5.178	0.560	5.030
	5.000	0.148	7.607	0.367	7.614
	10.000	0.070	14.887	0.180	15.230
	∞	1.511	1.768	1.455	0.673

n	R_L/R_S	L_1	C_2	L_3	C_4

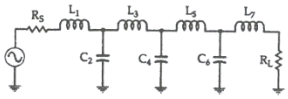


Chebyshev Low-Pass Prototype Element Values for 0.1-dB Ripple

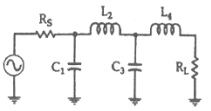


n	R_S/R_L	C_1	L_2	C_3	L_4	C_5	L_6	C_7
5	1.000	1.301	1.556	2.241	1.556	1.301		
	0.900	1.285	1.433	2.380	1.488	1.488		
	0.800	1.300	1.282	2.582	1.382	1.738		
	0.700	1.358	1.117	2.868	1.244	2.062		
	0.600	1.470	0.947	3.269	1.085	2.484		
	0.500	1.654	0.778	3.845	0.913	3.055		
	0.400	1.954	0.612	4.720	0.733	3.886		
	0.300	2.477	0.451	6.196	0.550	5.237		
	0.200	3.546	0.295	9.127	0.366	7.889		
	0.100	6.787	0.115	17.957	0.182	15.745		
	∞	1.561	1.807	1.766	1.417	0.651		
6	1.355	0.942	2.080	1.659	2.247	1.534	1.277	
	1.429	0.735	2.249	1.454	2.544	1.405	1.629	
	1.667	0.542	2.600	1.183	3.064	1.185	2.174	
	2.000	0.414	3.068	0.958	3.712	0.979	2.794	
	2.500	0.310	3.765	0.749	4.651	0.778	3.645	
	3.333	0.220	4.927	0.551	6.195	0.580	4.996	
	5.000	0.139	7.250	0.361	9.261	0.384	7.618	
	10.000	0.067	14.220	0.178	18.427	0.190	15.350	
	∞	1.534	1.884	1.831	1.749	1.394	0.638	
7	1.000	1.262	1.520	2.239	1.680	2.239	1.520	1.262
	0.900	1.242	1.395	2.361	1.578	2.397	1.459	1.447
	0.800	1.255	1.245	2.548	1.443	2.624	1.362	1.697
	0.700	1.310	1.083	2.819	1.283	2.942	1.233	2.021
	0.600	1.417	0.917	3.205	1.209	3.384	1.081	2.444
	0.500	1.595	0.753	3.764	0.928	4.015	0.914	3.018
	0.400	1.885	0.593	4.618	0.742	4.970	0.738	3.855
	0.300	2.392	0.437	6.054	0.556	6.569	0.557	5.217
	0.200	3.428	0.286	8.937	0.369	9.770	0.372	7.890
	0.100	6.570	0.141	17.603	0.184	19.376	0.186	15.813
	∞	1.575	1.858	1.921	1.827	1.734	1.379	0.631

n	R_L/R_S	L_1	C_2	L_3	C_4	L_5	C_6	L_7

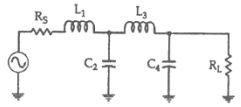


Chebyshev Low-Pass Prototype Element Values for 0.5-dB Ripple

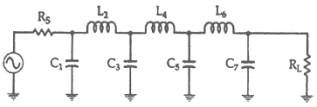


n	R_S/R_L	C_1	L_2	C_3	L_4
2	1.984	0.983	1.950		
	2.000	0.909	2.103		
	2.500	0.564	3.165		
	3.333	0.375	4.411		
	5.000	0.228	6.700		
	10.000	0.105	13.322		
	∞	1.307	0.975		
3	1.000	1.864	1.280	1.834	
	0.900	1.918	1.209	2.026	
	0.800	1.997	1.120	2.237	
	0.700	2.114	1.015	2.517	
	0.500	2.557	0.759	3.436	
	0.400	2.985	0.615	4.242	
	0.300	3.729	0.463	5.576	
	0.200	5.254	0.309	8.225	
	0.100	9.890	0.153	16.118	
	∞	1.572	1.518	0.932	
4	1.984	0.920	2.586	1.304	1.826
	2.000	0.845	2.720	1.238	1.985
	2.500	0.516	3.766	0.869	3.121
	3.333	0.344	5.120	0.621	4.480
	5.000	0.210	7.708	0.400	6.987
	10.000	0.098	15.352	0.194	14.262
	∞	1.436	1.889	1.521	0.913

n	R_L/R_S	L_1	C_2	L_3	C_4



Chebyshev Low-Pass Prototype Element Values for 0.5-dB Ripple



n	R_S/R_L	C_1	L_2	C_3	L_4	C_5	L_6	C_7
5	1.000	1.807	1.303	2.691	1.303	1.807		
	0.900	1.854	1.222	2.849	1.238	1.970		
	0.800	1.926	1.126	3.060	1.157	2.185		
	0.700	2.035	1.015	3.353	1.058	2.470		
	0.600	2.200	0.890	3.765	0.942	2.861		
	0.500	2.457	0.754	4.367	0.810	3.414		
	0.400	2.870	0.609	5.296	0.664	4.245		
	0.300	3.588	0.459	6.871	0.508	5.625		
	0.200	5.064	0.306	10.054	0.343	8.367		
	0.100	9.556	0.153	19.647	0.173	16.574		
	∞	1.630	1.740	1.922	1.514	0.903		
6	1.984	0.905	2.577	1.368	2.713	1.299	1.796	
	2.000	0.830	2.704	1.291	2.872	1.237	1.956	
	2.500	0.506	3.722	0.890	4.109	0.881	3.103	
	3.333	0.337	5.055	0.632	5.699	0.635	4.481	
	5.000	0.206	7.615	0.406	8.732	0.412	7.031	
	10.000	0.096	15.186	0.197	17.681	0.202	14.433	
7	1.000	1.790	1.296	2.718	1.385	2.718	1.296	1.790
	0.900	1.835	1.215	2.869	1.308	2.883	1.234	1.953
	0.800	1.905	1.118	3.076	1.215	3.107	1.155	2.168
	0.700	2.011	1.007	3.364	1.105	3.416	1.058	2.455
	0.600	2.174	0.882	3.772	0.979	3.852	0.944	2.848
	0.500	2.428	0.747	4.370	0.838	2.289	0.814	3.405
	0.400	2.835	0.604	5.295	0.685	5.470	0.669	4.243
	0.300	3.546	0.455	6.867	0.522	7.134	0.513	5.635
	0.200	5.007	0.303	10.049	0.352	10.496	0.348	8.404
	0.100	9.456	0.151	19.649	0.178	20.631	0.176	16.665
	∞	1.646	1.777	2.031	1.789	1.924	1.503	0.895

n	R_L/R_S	L_1	C_2	L_3	C_4	L_5	C_6	L_7

